

*MA.8.NSO.1.5***Benchmark****MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.**

Example: The sum of 2.31×10^{15} and 9.1×10^{13} is 2.401×10^{15} .

Benchmark Clarifications:

Clarification 1: Within this benchmark, for addition and subtraction with numbers expressed in scientific notation, exponents are limited to within 2 of each other.

Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.4, MA.8.NSO.1.5, MA.8.NSO.1.6

Terms from the K-12 Glossary

- Scientific Notation

Vertical Alignment**Previous Benchmarks**

- MA.7.NSO.1.1
- MA.7.NSO.2.1

Next Benchmarks

- MA.912.GR.4.5

Purpose and Instructional Strategies

In grade 7, students developed an understanding of the Laws of Exponents (Appendix E) with numerical expressions. They focused on generating equivalent numerical expressions with whole-number exponents and rational number bases. In grade 8, students use the knowledge of Laws of Exponents to work with scientific notation. In Geometry, students will solve problems involving density in terms of area and volume which can be represented using scientific notation when the numbers are large. Additionally, students can apply their scientific notation knowledge in science courses.

- Instruction connects scientific notation work with the Laws of Exponents with integer exponents.
- Instruction includes having students color code or use a highlighter to help keep the numbers together.
 - For example, when multiplying 3.2×10^{28} and 6.7×10^7 , students can highlight the 3.2 and 6.7 in one color and the 10^{28} and 10^7 in another color for organizational purposes.
- Students should develop fluency with and without using a calculator when performing operations with numbers expressed in scientific notation.
- It is helpful to include contextual problems to compare numbers written in scientific notation, including cross-curricular examples from science.

Common Misconceptions or Errors

- Some students may incorrectly apply addition and subtraction across a problem.
 - For example, students may miscalculate $(1.3 \times 10^3) + (3.4 \times 10^5)$ as 4.7×10^8 .

- Some students may incorrectly apply multiplication across a problem.
 - For example, students may miscalculate $(2 \times 10^4)(3 \times 10^5)$ as 6×10^{20} .
- Some students may incorrectly represent their final answer, not in scientific notation.
 - For example, students may write $(2 \times 10^4)(6 \times 10^5)$ as 12×10^9 instead of 1.2×10^{10} .

Strategies to Support Tiered Instruction

- Instruction includes making connections of a number written in standard form to the same number written in scientific notation. Key connections include recognizing the similarities in the first two digits of both numbers and the connections between the place value of the number in standard form and the exponent of the power.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.6.1)

A collection of meteorites includes three meteorites that weigh 1.1×10^2 grams, 6.8×10^2 grams, and 8.4×10^{-2} grams.

Part A. Why would a scientist represent the weights using scientific notation? Are all the meteorites approximately the same size?

Part B. What is the difference between the mass of the heaviest meteorite and the mass of the lightest meteorite? Write your answer in standard notation.

Instructional Items

Instructional Item 1

What is the sum of 7×10^{-8} and 6×10^{-8} ?

Instructional Item 2

Write the expression shown as a number in scientific number.

$$\frac{(8 \times 10^2)(7.5 \times 10^4)}{5 \times 10^2}$$

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*